

06 — State & Lifecycle Specification

Product State Machines, Transition Rules, Lifecycle Gates & Trusted-State Semantics

Field	Value
Document ID	CloudShield AI — 06
Version	1.0
Status	Engineering Baseline
Scope	Product-wide state and lifecycle behavior
Primary State Layers	Observed, Proposed, Verified, Trusted
Change Lifecycle	Candidate →Impact →Verification →Governance →Promotion
DAIL Position	Deterministic lifecycle/safety subsystem within CloudShield AI

State is evidence-backed. Promotion is earned. Invalid state is explicit.

Document Contents

1. Purpose and Scope
2. Lifecycle Philosophy
3. State Taxonomy
4. Aggregate and State Ownership
5. Cloud Connection Lifecycle
6. Discovery Job Lifecycle
7. Asset Lifecycle
8. Observation Lifecycle
9. Configuration Snapshot Lifecycle
10. Finding Lifecycle
11. Risk Assessment Lifecycle
12. Dependency and Impact Lifecycle
13. AI Interaction Lifecycle
14. Remediation Candidate Lifecycle
15. DAIL Lifecycle
16. Verification Lifecycle
17. Governance and Approval Lifecycle
18. Promotion Lifecycle
19. Trusted State Lifecycle
20. Drift and Invalidation Lifecycle
21. Evidence and Audit Lifecycle
22. Global Transition Rules
23. Concurrency, Idempotency and Ordering
24. Failure and Recovery Semantics
25. State Reconciliation
26. MVP Lifecycle Slice
27. Acceptance Criteria
28. Traceability
29. Glossary

1. Purpose and Scope

This specification formalizes how CloudShield AI represents and transitions product state. It defines lifecycle states, legal transitions, transition triggers, guards, side effects, failure behavior, recovery semantics and trusted-state rules across the product.

The purpose is to prevent ambiguous state and unsafe progression. In particular, observed cloud state, AI-generated recommendations, proposed infrastructure state, verified state and trusted state must never be conflated.

1.1 Scope

- Product-wide lifecycle semantics.
- State machines for core domain aggregates.
- DAIL-controlled change lifecycle.
- Verification and promotion gates.
- Drift/invalidation behavior.
- Concurrency and idempotency.
- Failure/recovery and reconciliation.

2. Lifecycle Philosophy

Principle	Required Behavior
Explicit state	Every important aggregate has a defined finite or controlled lifecycle.
Legal transitions only	A state change must satisfy explicit preconditions.
Evidence-backed	Material transitions produce traceable evidence/audit records.
Fail closed	Safety-critical uncertainty blocks progression instead of assuming success.
No hidden mutation	Side effects occur only through explicit transitions/operations.
Trusted state is earned	Only successful governed promotion can advance trusted state.
Observed is independent	Provider observations do not overwrite trusted state.
Version-bound	Verification/promotion apply to exact candidate versions.
Idempotent	Retries do not duplicate irreversible transitions.
Recoverable	Transient failures can resume/reconcile without corrupting state.

Canonical control loop: DISCOVER →INVENTORY →ANALYZE →FINDING →RISK →AI →CANDIDATE →IMPACT →VERIFY →DECIDE →PROMOTE →TRUSTED STATE →MONITOR.

3. State Taxonomy

3.1 Four Product State Layers

Layer	Definition	Authority
OBSERVED	Latest provider-reported reality captured by CloudShield AI.	Cloud provider observation
PROPOSED	Intended state represented by a candidate/recommendation.	Human/AI/rule proposal
VERIFIED	State confirmed by deterministic	Verification subsystem

	verification.	
TRUSTED	State accepted after required governance and promotion.	Governance + promotion

STATE-001: The system shall maintain explicit separation among observed, proposed, verified and trusted state.

STATE-002: A state layer shall not implicitly inherit authority from another layer.

3.2 State Qualifiers

- FRESH / STALE
- COMPLETE / PARTIAL
- VALID / INVALIDATED
- ACTIVE / SUPERSEDED
- PASS / FAIL / BLOCKED
- AUTHORIZED / UNAUTHORIZED

Qualifiers may be represented as state fields or derived status, but their semantics must remain explicit.

4. Aggregate and State Ownership

Aggregate	State Owner	Primary Transition Authority
CloudConnection	Cloud Connectivity	Connection workflow/service
DiscoveryJob	Operations/Discovery	Job worker/orchestrator
Asset	Inventory	Inventory reconciliation
Finding	Security	Analysis/reconciliation
RiskAssessment	Risk	Risk calculation
ImpactAnalysis	DAIL/Impact	Impact engine
AIRequest/Response	AI	AI orchestration
Candidate	Change Management	Candidate lifecycle + DAIL gates
VerificationRun	Verification	Verification engine
Approval/Decision	Governance	Authorized decision actor/policy
Promotion	DAIL/Promotion	Promotion controller
TrustedState	Trusted State	Promotion controller
Evidence/Audit	Evidence	Append-only event generation

STATE-OWN-001: Only the owning domain or explicitly authorized lifecycle service shall mutate aggregate state.

5. Cloud Connection Lifecycle

State	Meaning
CONFIGURED	Connection data exists but required validation is incomplete.
VALIDATING	Credential/permission validation is in progress.
READY	Required access has been validated.
DEGRADED	Connection works only partially or scope/permissions are incomplete.
DISABLED	Connection intentionally disabled.
ERROR	Connection cannot perform required operations.

From	Trigger	To	Guard
CONFIGURED	Start validation	VALIDATING	Authorized
VALIDATING	Validation passes	READY	Required permissions pass
VALIDATING	Partial capability	DEGRADED	Policy permits degraded

			operation
VALIDATING	Validation fails	ERROR	Failure recorded
READY	Material capability loss	DEGRADED	Observed by validation/operation
READY	Disable	DISABLED	Authorized
DEGRADED	Successful revalidation	READY	Required scope restored
ERROR	Successful revalidation	READY	Required validation passes
DISABLED	Enable + validate	VALIDATING	Authorized

STATE-CONN-001: Discovery shall not start against a connection unless its state satisfies the configured discovery precondition.

6. Discovery Job Lifecycle

State	Meaning
QUEUED	Accepted for execution.
RUNNING	Worker is actively executing.
SUCCEEDED	Completed with required scope.
PARTIAL	Completed with explicitly recorded incomplete scope/results.
FAILED	Terminal failure.
CANCELLED	Stopped by authorized/system action.

Transition	Rule
QUEUED → RUNNING	Worker claims job using lease/idempotency control.
RUNNING → SUCCEEDED	Required operations complete.
RUNNING → PARTIAL	Some scope failed but job produces valid partial results.
RUNNING → FAILED	Required operation fails without safe continuation.
RUNNING → CANCELLED	Cancellation accepted and worker stops safely.
FAILED/PARTIAL → QUEUED	Explicit retry/reconciliation creates a new attempt or resumes where safe.

STATE-JOB-001: Job retries shall not duplicate irreversible domain transitions.

STATE-JOB-002: Final job status shall be durable independently of queue delivery.

7. Asset Lifecycle

State	Meaning
DISCOVERED	Asset has been observed for the first time.
ACTIVE	Asset continues to be observed within expected scope.
STALE	Asset has not been observed within freshness policy.
MISSING	Expected asset is no longer observed.
RETIRED	Asset is determined to have left the managed scope.
CONFLICTED	Identity or reconciliation ambiguity prevents safe merge.

From	Trigger	To
DISCOVERED	Successful reconciliation	ACTIVE
ACTIVE	Fresh observation	ACTIVE
ACTIVE	Freshness threshold exceeded	STALE
STALE	Observation returns	ACTIVE
STALE	Confirmed absent	MISSING

MISSING	Scope retirement/confirmation	RETIRED
ANY	Identity conflict	CONFLICTED
CONFLICTED	Human/deterministic resolution	ACTIVE

Asset lifecycle does not imply that a missing asset was deleted by CloudShield AI; it represents product observation status.

8. Observation Lifecycle

Observations are immutable point-in-time records. Their lifecycle concerns collection completeness and processing, not mutation of the historical fact.

Collection Status	Meaning
COLLECTED	Provider data captured.
NORMALIZED	Converted to canonical representation.
PARTIAL	Collection incomplete.
INVALID	Failed schema/semantic validation.
PROCESSED	Applied to relevant product aggregates.

STATE-OBS-001: Historical observations shall not be overwritten to reflect later observations.

STATE-OBS-002: Invalid or partial observations shall remain distinguishable from complete observations.

9. Configuration Snapshot Lifecycle

State	Meaning
DRAFT	Being assembled from observation data.
FINAL	Immutable normalized snapshot.
SUPERSEDED	No longer the latest snapshot for the asset.
INVALIDATED	Known assumptions make the snapshot unsuitable for a referenced decision.

Transition	Guard
DRAFT → FINAL	Schema, provenance and content validation pass.
FINAL → SUPERSEDED	A newer valid snapshot becomes current.
FINAL → INVALIDATED	Relevant external change invalidates its use in a decision.
SUPERSEDED → INVALIDATED	Historical snapshot later determined unsuitable for a referenced decision.

STATE-CONF-001: Final configuration snapshots shall be immutable.

STATE-CONF-002: Snapshot invalidation shall not delete the underlying snapshot.

10. Finding Lifecycle

State	Meaning
OPEN	Condition currently detected.
ACKNOWLEDGED	Reviewed but not resolved.
IN_REMEDIATION	Candidate is actively associated.
VERIFIED	Corrective condition verified.

RESOLVED	Finding recognized as resolved.
REOPENED	Condition returned.
SUPPRESSED	Authorized suppression is active.

From	Trigger	To	Guard
OPEN	Review	ACKNOWLEDGED	Authorized user
OPEN/ACKNOWLEDGED	Candidate linked	IN_REMEDIATION	Candidate valid
IN_REMEDIATION	Verification passes	VERIFIED	Relevant check passes
VERIFIED	Resolution confirmed	RESOLVED	Current observation supports resolution
RESOLVED	Condition returns	REOPENED	Fresh detection
OPEN/ACKNOWLEDGED	Suppress	SUPPRESSED	Policy + reason
SUPPRESSED	Suppression removed/expired	OPEN	Re-evaluation

STATE-FIND-001: A finding shall not be marked resolved solely because a candidate was created or approved.

11. Risk Assessment Lifecycle

State	Meaning
CALCULATING	Risk computation is running.
CALCULATED	Valid result exists.
STALE	Relevant inputs changed or freshness expired.
INVALIDATED	Inputs are no longer suitable for this result.

Transition	Rule
CALCULATING → CALCULATED	Required inputs and calculation succeed.
CALCULATED → STALE	Material input changes or freshness policy.
STALE → CALCULATING	Recalculation requested.
CALCULATED/STALE → INVALIDATED	Input provenance becomes unusable.

STATE-RISK-001: Risk results shall be refreshed or marked stale when material underlying context changes.

12. Dependency and Impact Lifecycle

Impact State	Meaning
NOT_STARTED	No analysis exists.
ANALYZING	Dependency traversal and policy analysis running.
PASSED	Impact is within acceptable policy bounds.
BLOCKED	Safety/policy condition prevents progression.
INCOMPLETE	Dependency data is insufficient for a reliable result.
SUPERSEDED	A newer analysis replaces this result.

From	Trigger	To
NOT_STARTED	Start	ANALYZING
ANALYZING	Safe result	PASSED
ANALYZING	Unsafe policy condition	BLOCKED
ANALYZING	Insufficient graph/context	INCOMPLETE
PASSED/BLOCKED/INCOMPLETE	New candidate version/context	SUPERSEDED
INCOMPLETE	Context restored	ANALYZING

STATE-IMP-001: An incomplete impact analysis shall not be treated as a passed safety gate.

13. AI Interaction Lifecycle

State	Meaning
REQUESTED	Structured request accepted.
CONTEXT_ASSEMBLING	Authorized context being assembled.
GENERATING	Provider invocation active.
RECEIVED	Provider response received.
VALIDATED	Response passes schema/semantic validation.
REJECTED	Response invalid or unsafe for downstream use.
EXPIRED	Response no longer valid for the referenced context.

Transition	Guard
REQUESTED → CONTEXT_ASSEMBLING	Authorized context request.
CONTEXT_ASSEMBLING → GENERATING	Context complete and policy-safe.
GENERATING → RECEIVED	Provider returns response.
RECEIVED → VALIDATED	Schema/semantic checks pass.
RECEIVED → REJECTED	Malformed/unsafe output.
VALIDATED → EXPIRED	Referenced context materially changed or TTL expired.

STATE-AI-001: Only VALIDATED AI output may be used to create a candidate through automated product workflow.

STATE-AI-002: AI output must become stale/expired when material source context changes according to policy.

14. Remediation Candidate Lifecycle

State	Meaning
DRAFT	Candidate is being created/edited.
READY_FOR_IMPACT	Candidate is complete enough for impact analysis.
IMPACTED	Impact analysis completed.
AWAITING_VERIFICATION	Candidate passed impact gate and awaits verification.
VERIFYING	Verification executing.
VERIFIED	Technical verification passed.
APPROVAL_REQUIRED	Governance decision required.
APPROVED	Governance approval obtained.
PROMOTING	Promotion executing.
PROMOTED	Trusted state advanced.
REJECTED	Governance rejected candidate.
BLOCKED	Technical/policy gate blocks progression.
FAILED	Execution/verification/promotion failed.
SUPERSEDED	Candidate version replaced by newer candidate.

From	Trigger	To	Required Gate
DRAFT	Complete	READY_FOR_IMPACT	Schema + authorization
READY_FOR_IMPACT	Impact run	IMPACTED	Impact result valid
READY_FOR_IMPACT	Unsafe/incomplete	BLOCKED	Fail closed
IMPACTED	Proceed	AWAITING_VERIFICATION	Impact PASSED
AWAITING_VERIFICATION	Start	VERIFYING	Candidate immutable
VERIFYING	Pass	VERIFIED	All blocking checks pass
VERIFYING	Fail	FAILED	Any blocking check fails
VERIFIED	Policy requires approval	APPROVAL_REQUIRED	Governance policy

VERIFIED	Auto approval permitted	APPROVED	Policy + technical gates
APPROVAL_REQUIRED	Approve	APPROVED	Authorized approver
APPROVAL_REQUIRED	Reject	REJECTED	Authorized decision
APPROVED	Promote	PROMOTING	Preconditions rechecked
PROMOTING	Success	PROMOTED	Post-change confirmation
PROMOTING	Failure	FAILED	Trusted state unchanged

STATE-CAND-001: Candidate content shall become immutable before verification begins.

STATE-CAND-002: A candidate cannot transition to PROMOTED unless all required technical and governance gates pass.

STATE-CAND-003: A candidate in FAILED, REJECTED or BLOCKED state shall remain historically traceable.

15. DAIL Lifecycle

DAIL is the deterministic lifecycle subsystem responsible for the controlled progression of infrastructure change candidates.

DAIL Gate	Functional State	Purpose
Identity	IDENTIFIED	Bind lifecycle to exact candidate/version.
Dependencies	ASSESSED	Resolve relevant relationships.
Impact	PASSED/BLOCKED/INCOMPLETE	Determine blast radius and invalidation.
Verification	PASSED/FAILED	Confirm intended post-change state.
Governance	DECIDED	Record approval/rejection.
Promotion	PROMOTED/FAILED	Establish or reject trusted-state transition.

STATE-DAIL-001: DAIL gates shall be evaluated in a deterministic, traceable sequence appropriate to the candidate.

STATE-DAIL-002: DAIL shall not treat AI output as a lifecycle gate pass.

STATE-DAIL-003: A failed gate shall prevent downstream promotion unless an explicitly governed alternate path exists.

16. Verification Lifecycle

State	Meaning
QUEUED	Verification requested.
RUNNING	Checks executing.
PASSED	All blocking checks passed.
FAILED	At least one blocking check failed.
PARTIAL	Some checks unavailable/incomplete; not equivalent to pass.
CANCELLED	Verification stopped.

16.1 Verification Gate Semantics

- PASS means all required blocking checks passed.

- FAIL means one or more required checks failed.
- PARTIAL means the evidence is insufficient to claim complete verification.
- CANCELLED means no conclusion of success is implied.
- Verification is bound to the exact candidate version and relevant environment/context.

STATE-VER-001: PARTIAL verification shall not satisfy a full verification gate.

STATE-VER-002: Verification results shall be immutable after completion.

17. Governance and Approval Lifecycle

Approval State	Meaning
NOT_REQUIRED	Policy does not require human approval.
REQUESTED	Approval request created.
UNDER_REVIEW	Approver is reviewing.
APPROVED	Authorized approval recorded.
REJECTED	Authorized rejection recorded.
ESCALATED	Different authority is required.
EXPIRED	Approval no longer valid due to age/context change.

Transition	Guard
REQUESTED → UNDER_REVIEW	Authorized approver opens request.
UNDER_REVIEW → APPROVED	Required evidence/gates pass and approver authorized.
UNDER_REVIEW → REJECTED	Approver rejects with required reason.
UNDER_REVIEW → ESCALATED	Policy/role requires higher authority.
APPROVED → EXPIRED	Material context/candidate version change or approval TTL.
NOT_REQUIRED → APPROVED	System records policy-based automatic approval where allowed.

STATE-GOV-001: An approval shall be invalidated when the exact candidate version or required safety context changes materially.

STATE-GOV-002: Approval shall not substitute for technical verification.

18. Promotion Lifecycle

State	Meaning
ELIGIBLE	All required gates have passed.
PROMOTING	Trusted-state transition executing.
SUCCEEDED	Promotion completed and trusted state advanced.
FAILED	Promotion failed; trusted state not advanced.
RECONCILIATION_REQUIRED	Outcome is ambiguous and must be reconciled before trust changes.

18.1 Promotion Preconditions

- Exact candidate version is identified.
- Impact gate is valid and passed.
- Verification is passed and not stale.
- Required governance decision is valid and not expired.
- No policy-defined invalidation has occurred.
- Required evidence can be durably recorded.

- Execution identity is authorized.

STATE-PROM-001: Promotion shall re-check critical preconditions immediately before the irreversible transition.

STATE-PROM-002: Ambiguous promotion outcomes shall enter RECONCILIATION_REQUIRED rather than claiming success.

STATE-PROM-003: Trusted state shall advance only after successful promotion confirmation.

19. Trusted State Lifecycle

State	Meaning
ACTIVE	Current trusted state.
SUPERSEDED	Replaced by a newer trusted state.
INVALIDATED	No longer safe to rely on due to material change.
UNKNOWN	Trusted state cannot currently be established/validated.

Transition	Trigger
ACTIVE →SUPERSEDED	Successful governed promotion creates newer trusted state.
ACTIVE →INVALIDATED	Material drift or dependency invalidation.
ACTIVE →UNKNOWN	Critical evidence/context becomes unavailable.
UNKNOWN →ACTIVE	New successful governed promotion establishes valid trust.
INVALIDATED →ACTIVE	Only through a new successful promotion; never by simply relabeling old state.

STATE-TRUST-001: Trusted state cannot be restored by changing a status field without a valid evidence-backed transition.

STATE-TRUST-002: Trusted-state history shall remain queryable after supersession or invalidation.

20. Drift and Invalidation Lifecycle

Drift occurs when current observed state differs materially from trusted or expected state. Invalidation occurs when a previously valid conclusion, verification or trust assumption can no longer be relied upon.

Condition	Result
Observed ≠ Trusted	Record DRIFT; evaluate invalidation policy.
Dependency changed	Re-evaluate impact and affected assumptions.
Verification context changed	Mark verification stale/invalidated.
Candidate content changed	Invalidate prior verification/approval.
Cloud scope/permissions changed	Potentially invalidate completeness-dependent conclusions.
Evidence unavailable	Mark dependent decisions/evidence as unavailable or unknown.

Drift State	Meaning
NOT_DETECTED	Observed and trusted state aligned or no comparison available.
DETECTED	Material difference identified.
ANALYZING	Impact/invalidation being evaluated.
ACKNOWLEDGED	User reviewed drift.
REMEDIATION_PENDING	Candidate associated.
RESOLVED	Observed state restored to accepted condition.

STATE-DRIFT-001: Drift shall not automatically trigger an infrastructure mutation.

STATE-DRIFT-002: Material drift shall be able to invalidate relevant trusted assumptions according to policy.

21. Evidence and Audit Lifecycle

Evidence State	Meaning
CAPTURED	Evidence created.
VALIDATED	Evidence schema/integrity validated.
LINKED	Associated with domain object/transition.
RETAINED	Within retention policy.
EXPIRED	Retention period ended.
QUARANTINED	Evidence integrity/classification issue prevents normal use.

STATE-EVID-001: Critical lifecycle transitions shall not be considered complete until required evidence persistence succeeds or the system enters an explicit reconciliation/error state.

STATE-EVID-002: Audit events shall be append-only from the product application perspective.

22. Global Transition Rules

GLOBAL-01: No transition without a valid current state.

GLOBAL-02: No transition may skip a mandatory gate.

GLOBAL-03: No successful transition may be emitted before required side effects are confirmed.

GLOBAL-04: Every material transition carries actor/service identity and correlation metadata.

GLOBAL-05: Every irreversible transition requires idempotency protection.

GLOBAL-06: State changes are atomic with the state record update wherever possible.

GLOBAL-07: External side effects and internal state changes require reconciliation when atomicity cannot be guaranteed.

GLOBAL-08: A stale prerequisite cannot satisfy a current safety gate.

GLOBAL-09: A lower-authority state cannot imply a higher-authority state.

GLOBAL-10: AI-generated state is advisory until transformed into a governed candidate and verified.

GLOBAL-11: Unknown/partial states must not satisfy completeness-sensitive gates.

GLOBAL-12: Historical states are never silently deleted to simplify the current view.

23. Concurrency, Idempotency and Ordering

23.1 Concurrency

- Candidate version changes must use optimistic locking or equivalent version checks.
- Only one active promotion should be allowed for a candidate/version.
- Approval and verification should be invalidated when protected content changes.
- Asset reconciliation must handle duplicate observations safely.

23.2 Idempotency

- Discovery result processing uses observation/job identity.
- Candidate transitions use candidate ID + version + transition/event identity.
- Verification runs use unique execution IDs.
- Promotion uses an idempotency key tied to candidate version and promotion attempt.
- Repeated event delivery must not create duplicate trusted states.

23.3 Ordering

- State transitions for one aggregate must preserve causal ordering.
- Out-of-order events are rejected, buffered or reconciled according to aggregate policy.
- A newer observation should not be overwritten by an older observation.

STATE-CONC-001: Concurrent lifecycle operations shall not allow two conflicting transitions to both become authoritative.

24. Failure and Recovery Semantics

Failure	Required State Behavior
Worker crash	Job remains recoverable/leased; retry without duplicate irreversible transition.
Queue redelivery	Idempotent processing.
Provider timeout	Operation remains incomplete/retryable; no false success.
Verification timeout	Verification remains non-passed until completed.
Approval service unavailable	Candidate remains awaiting decision.
Promotion API timeout	Enter reconciliation if external outcome is ambiguous.
Evidence store unavailable	Block safety-critical completion if evidence is mandatory.
Database conflict	Reject stale transition and require refresh/retry.
Permission loss	Mark affected operations degraded/failed and reassess validity.
Dependency graph incomplete	Impact becomes INCOMPLETE/BLOCKED, not PASSED.

24.1 Recovery Rule

Recovery should reconstruct authoritative state from durable records and external observations rather than assuming the last in-memory workflow step completed.

25. State Reconciliation

Reconciliation is required when internal state and external reality may disagree, especially around cloud mutations and promotion.

30. Identify the affected aggregate and operation ID.

31. Load the last authoritative internal state.
32. Query external provider state where safe and authorized.
33. Compare external observation with expected candidate/trusted state.
34. Determine whether the intended change occurred, did not occur, or is ambiguous.
35. Record reconciliation evidence.
36. Advance, fail, invalidate or quarantine state according to deterministic rules.
37. Never mark trusted state successful solely because an external call returned an ambiguous result.

STATE-REC-001: Ambiguous external outcomes shall enter an explicit reconciliation state.

STATE-REC-002: Reconciliation results shall be auditable and tied to the original operation.

26. MVP Lifecycle Slice

The MVP should implement a complete lifecycle for a constrained set of AWS resources rather than many partially defined states.

- Cloud connection: CONFIGURED →VALIDATING →READY/DEGRADED/ERROR.
- Discovery: QUEUED →RUNNING →SUCCEEDED/PARTIAL/FAILED.
- Asset: DISCOVERED →ACTIVE →STALE/MISSING/RETIRED.
- Finding: OPEN →ACKNOWLEDGED →IN_REMEDIATION →VERIFIED →RESOLVED.
- Candidate: DRAFT →READY_FOR_IMPACT →IMPACTED →VERIFYING →VERIFIED →APPROVAL_REQUIRED/APPROVED →PROMOTING →PROMOTED.
- DAIL: identity →dependency →impact →verification →governance →promotion.
- Trusted state: ACTIVE →SUPERSEDED/INVALIDATED.
- Drift: detection →analysis →remediation →resolution.
- Evidence: capture →validate/link →retained.
- All safety-critical transitions use idempotency and audit evidence.

27. Acceptance Criteria

LC-ACC-01: Every core aggregate has an explicit lifecycle or immutable-state semantics.

LC-ACC-02: Legal state transitions are enforced rather than represented only in UI.

LC-ACC-03: Observed, proposed, verified and trusted states are separate.

LC-ACC-04: Candidate content is immutable before verification.

LC-ACC-05: Verification is bound to an exact candidate version.

LC-ACC-06: Partial/incomplete verification cannot satisfy a full verification gate.

LC-ACC-07: Approval is invalidated when protected candidate/context changes.

LC-ACC-08: Promotion re-checks critical preconditions.

LC-ACC-09: Promotion failure does not advance trusted state.

LC-ACC-10: Ambiguous promotion outcomes enter reconciliation.

LC-ACC-11: Trusted state can be invalidated by material drift/dependency change.

LC-ACC-12: Invalidated trusted state cannot simply be relabeled active.

LC-ACC-13: Material transitions create evidence/audit records.

LC-ACC-14: Retries do not create duplicate irreversible transitions.

LC-ACC-15: Out-of-order events cannot overwrite newer authoritative state.

LC-ACC-16: Provider observations remain historically traceable.

LC-ACC-17: AI output cannot directly become trusted state.

LC-ACC-18: Incomplete dependency context cannot satisfy a safety gate.

LC-ACC-19: Recovery can reconstruct state from durable records and provider observations.

LC-ACC-20: The MVP lifecycle demonstrates the complete discover-to-trusted-state loop.

28. Traceability

Lifecycle Area	Domain/Data Alignment	Functional/UX Alignment
Cloud connection	CloudConnection	Onboarding and connection UX
Discovery	Job, Observation	Discovery workflow/job center
Asset	Asset	Inventory experience
Configuration	Snapshot/Diff	Configuration/drift UX
Finding	Finding	Finding lifecycle/detail
Risk	RiskAssessment	Risk prioritization
Dependency/Impact	DependencyEdge/ImpactAnalysis	Impact view
AI	AIRequest/Response/Recommendation	AI insight workflow
Candidate	Candidate/CandidateVersion	Change management UX
DAIL	DAILIdentity/LifecycleGate	DAIL lifecycle view
Verification	VerificationRun	Verification UX
Governance	Approval/Decision	Approval UX
Promotion	Promotion	Promotion confirmation
Trusted State	TrustedState	Trusted/drift experience
Evidence	EvidenceArtifact/AuditEvent	Evidence timeline

This lifecycle specification is the authoritative state-machine layer over the domain model defined in Document 05. Subsequent identity/dependency, security, AI and infrastructure specifications should use these transition semantics.

29. Glossary

Term	Definition
Aggregate	Domain object or cluster whose state transitions must remain consistent.
Observed State	Latest provider-reported state captured by CloudShield AI.
Proposed State	Intended state represented by a candidate.
Verified State	State confirmed by deterministic checks.
Trusted State	State accepted after governed promotion.
Transition	Authorized movement from one lifecycle state to another.
Gate	Precondition that must pass before a lifecycle transition.
Invalidation	Loss of validity of a previously usable result or assumption.
Drift	Material difference between observed and trusted/expected state.

Reconciliation	Process for resolving disagreement between internal state and external reality.
Idempotency	Property that repeating an operation does not duplicate its irreversible effect.
DAIL	Deterministic AI Infrastructure Lifecycle subsystem.
Promotion	Governed action that establishes a new trusted state.

Document Completion Statement

CloudShield AI State & Lifecycle Specification v1.0 establishes the authoritative lifecycle semantics for the product. It formalizes the distinction between observed, proposed, verified and trusted state and defines the controlled path from remediation candidate through DAIL, verification, governance and promotion.

Next planned document: 07 — CloudShield AI Identity & Dependency Specification.